



▲ FAIRNESS IMPROVED

● NO ACCURACY LOSS

Executive Summary

Fairness Experiment Report - Upload 31 summarizes the current fairness state for upload 31. Accuracy changed to +0.647 and fairness score changed to +0.985. Strong fairness improvement with no measurable accuracy loss. The best balanced model is ThresholdOptimizer_mitigated_threshold with combined score 0.850.

ACCURACY

0.647
▲ +0.647

FAIRNESS SCORE

0.985
▲ +0.985

COMBINED SCORE

0.850
ThresholdOptimizer_mitigated_threshold

DISPARATE IMPACT

0.943
Ratio (DIR)

Dataset & Model Information

DATASET	MODEL	SECTIONS INCLUDED
processed_dataset.csv	compas_log_reg_baseline.pkl	comparison

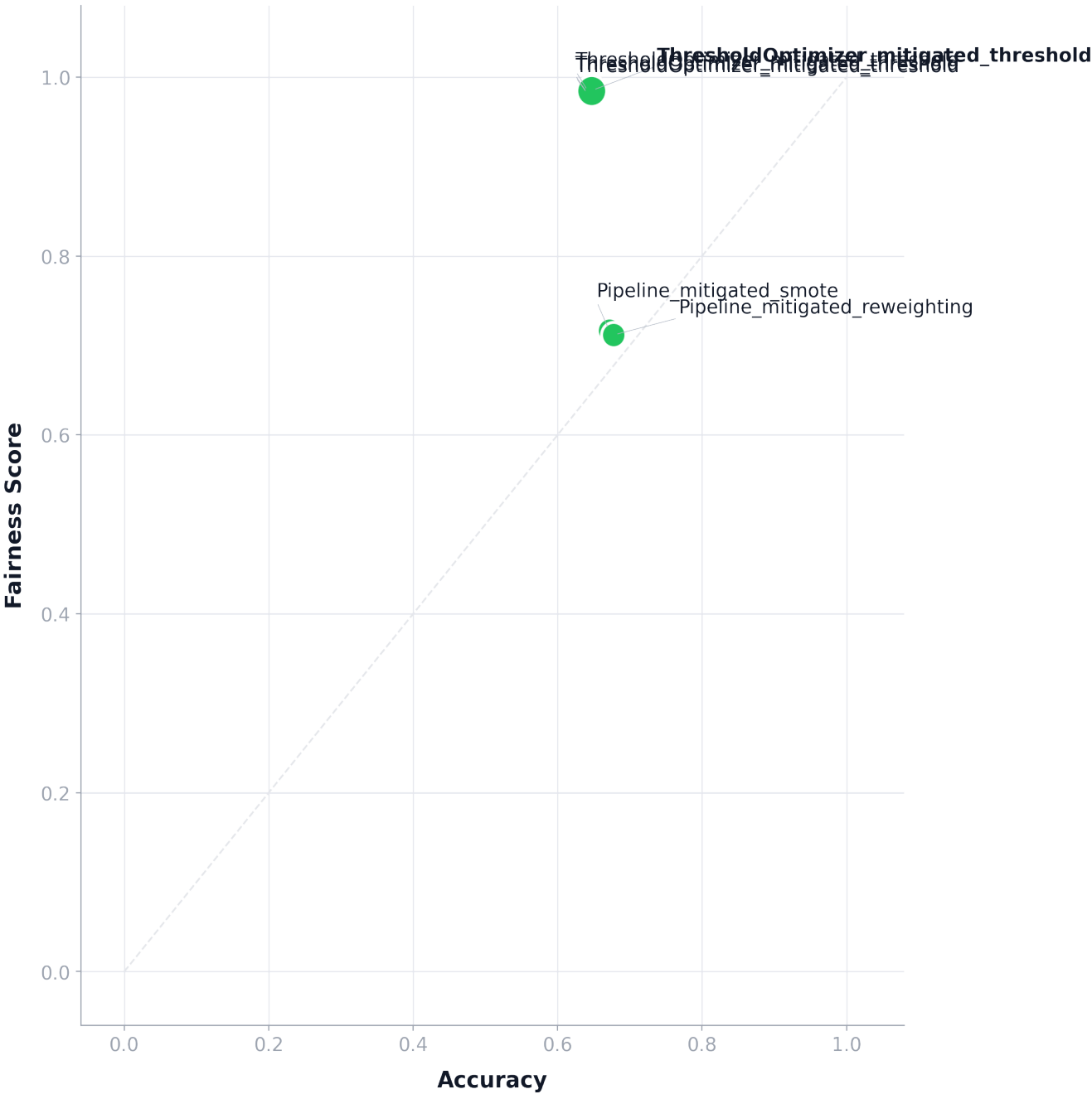
Fairness Metrics

METRIC	VALUE	INTERPRETATION
Demographic Parity Difference (DPD)	0.028	Difference in positive-prediction rates across groups; closer to 0 is more equitable.
Equalized Odds Difference (EOD)	0.002	Gap in true/false positive rates between groups; closer to 0 indicates equal error rates.
Disparate Impact Ratio (DIR)	0.943	Ratio of favorable outcome rates between groups; values near 1.0 indicate parity.

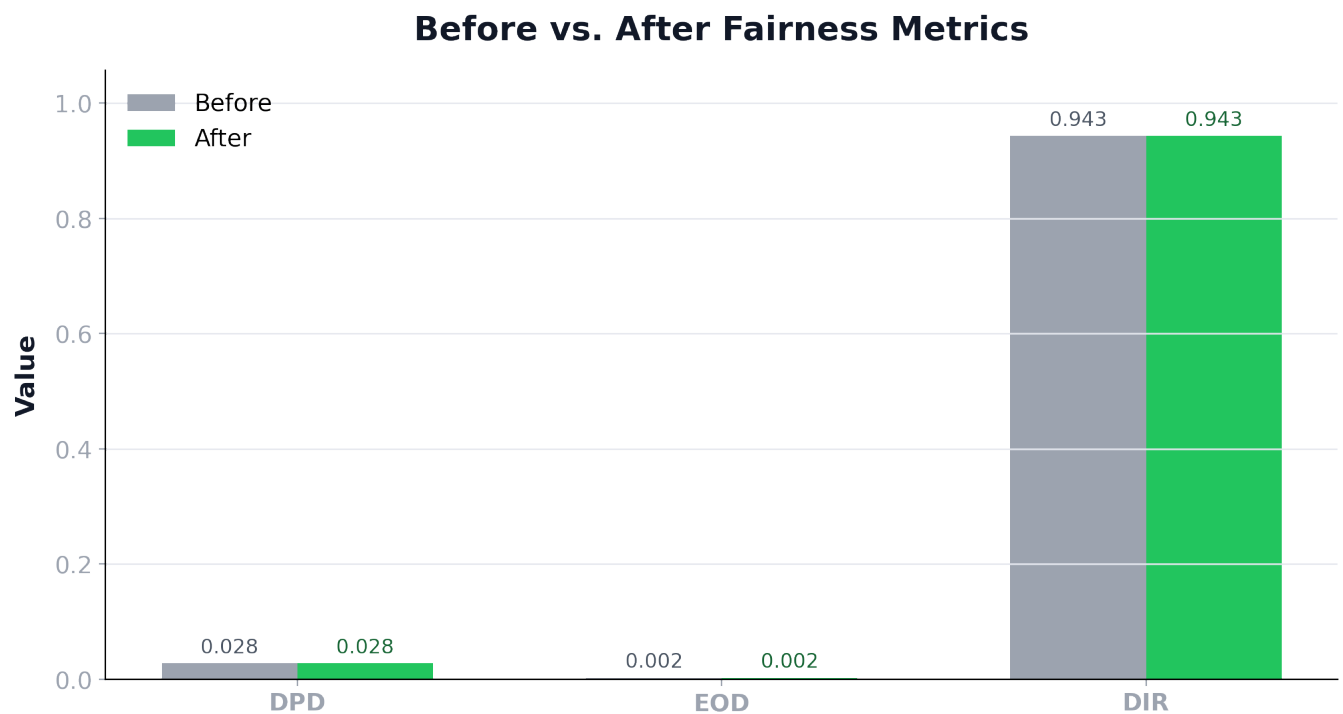
Model Comparison

Model	Accuracy	Fairness	DPD	EOD	DIR	Combined
★ ThresholdOptimizer_mitigated_threshold	0.647	0.985	0.028	0.002	0.943	0.850
★ ThresholdOptimizer_mitigated_threshold	0.643	0.984	0.024	0.009	0.951	0.847
★ ThresholdOptimizer_mitigated_threshold	0.641	0.982	0.033	0.003	0.933	0.846
★ ThresholdOptimizer_mitigated_threshold	0.642	0.981	0.036	0.003	0.928	0.845
Pipeline_mitigated_smote	0.671	0.717	0.274	0.293	0.534	0.699
Pipeline_mitigated_smote	0.671	0.717	0.274	0.293	0.534	0.699
Pipeline_mitigated_smote	0.671	0.717	0.274	0.293	0.534	0.699
Pipeline_mitigated_reweighting	0.677	0.712	0.274	0.302	0.479	0.698
Pipeline_mitigated_reweighting	0.677	0.712	0.274	0.302	0.479	0.698
Pipeline_mitigated_reweighting	0.677	0.712	0.274	0.302	0.479	0.698

Accuracy vs. Fairness Trade-off



Before vs. After Fairness Metrics



Recommendations

Fairness improved after mitigation. Consider validating and deploying the mitigated model.